

Route decision: Port of Shanghai → Port of Los Angeles

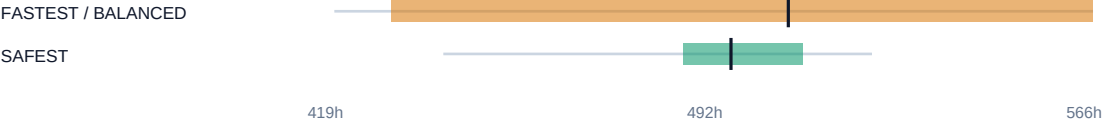
PORT-SHANGHAI → PORT-LOSANGELES · mode sea (STRICT) · cargo general · priority normal · scenarios: LA_PORT_STRIKE · planned 2026-09-29 13:47 UTC

Recommendation: FASTEST / BALANCED, SEA via direct. Expected arrival in **430 h** (17.9 days); plan for 507 h (p85), worst case 566 h (p95). Cost: freight S\$2,204. Peak leg risk 90%.

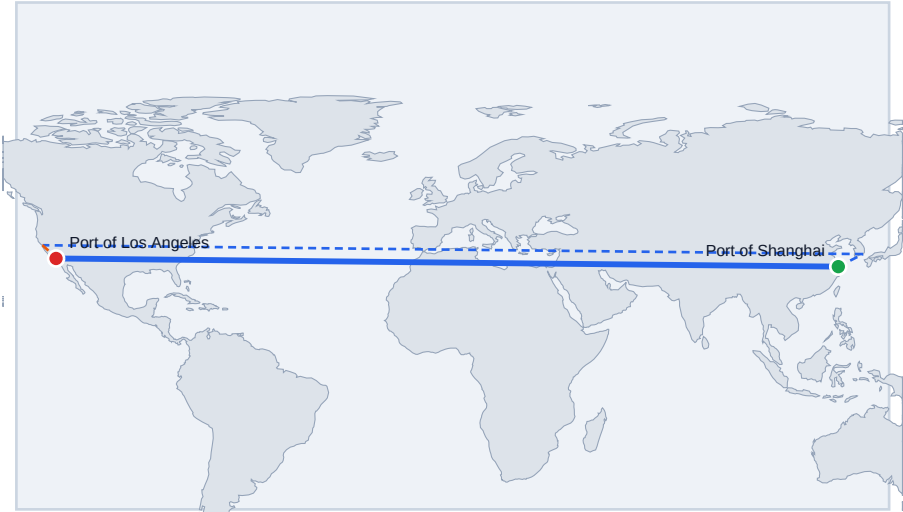
Options

Option	Mode	Via	Plan h	p50 h	p85 h	p95 h	Freight	Risk
FASTEST / BALANCED	SEA	direct	419	430	507	566	S\$2,204	90%
SAFEST	SEA	direct	440	486	496	509	S\$3,473	90%

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST / BALANCED: SEA via direct

SEA via no chokepoints; expected 429.8 h (p85 506.7 h, p95 565.7 h), S\$2,204 freight, 0 handoffs. 56.5 h faster than the next option. S\$1,269 cheaper (freight) than the lowest-cost alternative. Includes 120.0 h announced scenario delay.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	SEA	Port of Los Angeles	10,457	418.8	+88.0	S\$2,004	90%	SCENARIO: Terminal labor strike. Picket lines at all major berths. Throughput at 0%.

Transit 298.8 h + transfers 0.0 h + announced scenario delay 120.0 h = plan 418.8 h. Model delay buffer p50/p85/p95: +11.1 / +88.0 / +146.9 h. Route band p50 429.8 h, p85 506.7 h, p95 565.7 h (independent legs); 565.7 h if all legs are late together. Costs: linehaul S\$2,004, transfers S\$0.00, disruption surcharge S\$200.40.

Why the slowest leg (PORT-SHANGHAI → PORT-LOSANGELES, p85 208.0 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST: SEA via direct

SEA via no chokepoints; expected 486.3 h (p85 495.6 h, p95 509.4 h), S\$3,473 freight, 2 handoffs. 56.5 h slower than FASTEST. S\$1,269 more (freight) than FASTEST. Includes 120.0 h announced scenario delay.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	SEA	Port of Busan	824	23.5	+26.6	S\$157.94	0%	
2	SEA	Port of Oakland	9,080	259.4	+27.5	S\$1,740	0%	
3	handoff	Port of Oakland		14.0	+0.0	S\$319.42	8%	
4	ROAD	San Francisco Logistics Park	13	0.2	+3.6	S\$19.25	0%	
5	ROAD	LA Logistics Cluster	559	8.2	+3.6	S\$857.24	0%	
6	ROAD	Port of Los Angeles	35	120.5	+0.0	S\$53.96	90%	SCENARIO: Terminal labor strike. Picket lines at all major berths. Throughput at 0%.
7	handoff	Port of Los Angeles		14.0	+0.0	S\$319.42	8%	

Transit 291.9 h + transfers 28.0 h + announced scenario delay 120.0 h = plan 439.9 h. Model delay buffer p50/p85/p95: +46.4 / +61.4 / +83.0 h. Route band p50 486.3 h, p85 495.6 h, p95 509.4 h (independent legs); 522.9 h if all legs are late together. Costs: linehaul S\$2,829, transfers S\$638.83, disruption surcharge S\$5.39.

Why the slowest leg (PORT-BUSAN → PORT-OAKLAND, p85 27.5 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.